

PRACTICAL 4

The screenshot displays the SQL Enterprise Manager interface. The left pane shows the 'SCHEMAS' tree with a search filter 'Filter objects'. The main pane shows a SQL script with the following statements:

```
1 DROP DATABASE IF EXISTS bookflow_db;
2 CREATE DATABASE bookflow_db;
3 USE bookflow_db;
4
5 SELECT DATABASE();
```

The 'Result Grid' shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	23:10:45	DROP DATABASE IF EXISTS bookflow_db	0 row(s) affected, 1 warning(s): 1008 Can't drop dat...	0.015 sec
2	23:10:45	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
3	23:10:45	USE bookflow_db	0 row(s) affected	0.000 sec
4	23:11:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The bottom pane shows the 'Output' tab with 'Action Output' selected. The 'Object Info' tab shows 'No object selected'.

FileEditViewQueryDatabaseServerToolsScriptingHelp

SQL File 4*SQL File 3*

Limit to 1000 rows

SCHEMAS

Filter objects

practical1

practical2

practical4

skill3

AdministrationSchemas

Information

No object selected

Object InfoSession

1•DROP DATABASE IF EXISTS bookflow_db;

2•CREATE DATABASE bookflow_db;

3•USE bookflow_db;

4

5•SELECT DATABASE();

6

7•CREATE TABLE Books

8(

9book_id INT AUTO_INCREMENT PRIMARY KEY,

10title VARCHAR(200) NOT NULL,

11author VARCHAR(150) NOT NULL,

12genre VARCHAR(100),

13description TEXT,

14published_year INT

15);

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	23:10:45	DROP DATABASE IF EXISTS bookflow_db	0 row(s) affected, 1 warning(s): 1008 Can't drop dat...	0.015 sec
2	23:10:45	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
3	23:10:45	USE bookflow_db	0 row(s) affected	0.000 sec
4	23:11:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	23:12:31	CREATE TABLE Books (book_id INT AUTO_IN...	0 row(s) affected	0.031 sec

Filter objects

- practical1
- practical2
- practical4
- skill3

```

150 WHERE book_id = 10;
151
152 • SELECT
153     book_id,
154     title,
155     embedding
156 FROM Books
157 ORDER BY book_id;

```

Result Grid

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]

Books 5 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 11	23:16:44	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓ 12	23:17:07	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓ 13	23:17:36	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 14	23:17:48	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓ 15	23:17:57	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓ 16	23:18:09	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓ 17	23:18:20	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.015 sec
✓ 18	23:18:33	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓ 19	23:18:42	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓ 20	23:18:55	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
✓ 21	23:19:06	SELECT book_id, title, embedding FROM ...	10 row(s) returned	0.000 sec / 0.000 sec

Administration Schemas

Information

No object selected

Object Info Session

16

17 • DESCRIBE Books;

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

Result 2 x

Read Only

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- practical1
- practical2
- practical4
- skill2

SQL File 4*

```
19 • INSERT INTO Books
20 (
21     title,
22     author,
23     genre,
24     description,
25     published_year
26 )
27 VALUES
28 (
29     'The Martian',
30     'Andy Weir',
31     'Science Fiction',
32     'An astronaut struggles to survive alone on Mars.',
33     2011
34 ),
35
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	23:10:45	DROP DATABASE IF EXISTS bookflow_db	0 row(s) affected, 1 warning(s): 1008 Can't drop database 'bookflow_db': database doesn't exist	0.015 sec
2	23:10:45	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
3	23:10:45	USE bookflow_db	0 row(s) affected	0.000 sec
4	23:11:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	23:12:31	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200...	0 row(s) affected	0.031 sec
6	23:13:00	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
7	23:14:12	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.016 sec

Object Info Session

SQL File 4* SQL File 3*

Limit to 1000 rows

```
99 • SELECT
100     book_id,
101     title,
102     author,
103     genre,
104     published_year
105 FROM Books
106 ORDER BY book_id;
```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content: I A

	book_id	title	author	genre	published_year
▶	1	The Martian	Andy Weir	Science Fiction	2011
	2	Dune	Frank Herbert	Science Fiction	1965
	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
	4	Pride and Prejudice	Jane Austen	Romance	1813
	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
	6	Foundation	Isaac Asimov	Science Fiction	1951

Books 3 x

Result Grid

Form Editor

Apply

Output

Administration Schemas

Information

No object selected

Object Info Session

110

111 • DESCRIBE Books;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	

Result 4 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	23:10:45	DROP DATABASE IF EXISTS bookflow_db	0 row(s) affected, 1 warning(s): 1008 Can't drop dat...	0.015 sec
2	23:10:45	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
3	23:10:45	USE bookflow_db	0 row(s) affected	0.000 sec
4	23:11:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	23:12:31	CREATE TABLE Books (book_id INT AUTO_IN...	0 row(s) affected	0.031 sec
6	23:13:00	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
7	23:14:12	INSERT INTO Books (title, author, genre, ...	10 row(s) affected Records: 10 Duplicates: 0 War...	0.016 sec
8	23:14:58	SELECT book_id, title, author, genre, p...	10 row(s) returned	0.000 sec / 0.000 sec
9	23:15:28	ALTER TABLE Books ADD COLUMN embedding J...	0 row(s) affected Records: 0 Duplicates: 0 Wamin...	0.032 sec
10	23:16:07	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec

112

113 • UPDATE Books

114 SET embedding = JSON_ARRAY(0.90, 0.80, 0.20)

115 WHERE book_id = 1;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	23:10:45	DROP DATABASE IF EXISTS bookflow_db	0 row(s) affected, 1 warning(s): 1008 Can't drop dat...	0.015 sec
2	23:10:45	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
3	23:10:45	USE bookflow_db	0 row(s) affected	0.000 sec
4	23:11:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	23:12:31	CREATE TABLE Books (book_id INT AUTO_IN...	0 row(s) affected	0.031 sec
6	23:13:00	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
7	23:14:12	INSERT INTO Books (title, author, genre, ...	10 row(s) affected Records: 10 Duplicates: 0 War...	0.016 sec
8	23:14:58	SELECT book_id, title, author, genre, p...	10 row(s) returned	0.000 sec / 0.000 sec
9	23:15:28	ALTER TABLE Books ADD COLUMN embedding J...	0 row(s) affected Records: 0 Duplicates: 0 Wamin...	0.032 sec
10	23:16:07	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec
11	23:16:44	UPDATE Books SET embedding = JSON_ARRAY(...	1 row(s) affected Rows matched: 1 Changed: 1 W...	0.016 sec

Administration Schemas

Information

No object selected

Object Info Session

AdministrationSchemas

Information

No object selected

Object InfoSession

Navigator

SCHEMAS

Filter objects

practical1

practical2

practical4

skill3

117 • UPDATE Books

118 SET embedding = JSON_ARRAY(0.85, 0.75, 0.25)

119 WHERE book_id = 2;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	23:10:45	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
3	23:10:45	USE bookflow_db	0 row(s) affected	0.000 sec
4	23:11:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
5	23:12:31	CREATE TABLE Books (book_id INT AUTO_I...	0 row(s) affected	0.031 sec
6	23:13:00	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
7	23:14:12	INSERT INTO Books (title, author, genre,...	10 row(s) affected Records: 10 Duplicates: 0 Wa...	0.016 sec
8	23:14:58	SELECT book_id, title, author, genre, ...	10 row(s) returned	0.000 sec / 0.000 sec
9	23:15:28	ALTER TABLE Books ADD COLUMN embedding...	0 row(s) affected Records: 0 Duplicates: 0 Wami...	0.032 sec
10	23:16:07	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec
11	23:16:44	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
12	23:17:07	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec

SQL File 4*

SQL File 3*

Limit to 1000 rows

title,
author,
genre,
ROUND(
SQRT(
POW(
CAST(
JSON_EXTRACT(embedding, '\$[0]')
AS DECIMAL(10,6)

Result Grid

Filter Rows:

Export:

Wrap Cell Contents

Fetch rows:

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0
8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	Dune	Frank Herbert	Science Fiction	0.0866

Result 6 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	23:17:07	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
13	23:17:36	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
14	23:17:48	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
15	23:17:57	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
16	23:18:09	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
17	23:18:20	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.015 sec
18	23:18:33	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
19	23:18:42	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
20	23:18:55	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
21	23:19:06	SELECT book_id, title, embedding FROM ...	10 row(s) returned	0.000 sec / 0.000 sec
22	23:20:07	SELECT book_id, title, author, genre, ...	3 row(s) returned	0.015 sec / 0.000 sec

SCHEMAS

Filter objects

practical1

practical2

practical4

skill3

194

ORDER BY l2_distance ASC

195

LIMIT 3;

196

197

SELECT

198

book_id,

199

title,

200

author,

201

genre,

202

cosine_similarity

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Fetch rows:

Result Grid

Form Editor

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 7 ×

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	13 23:17:36	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	14 23:17:48	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓	15 23:17:57	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓	16 23:18:09	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓	17 23:18:20	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.015 sec
✓	18 23:18:33	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.000 sec
✓	19 23:18:42	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.016 sec
✓	20 23:18:55	UPDATE Books SET embedding = JSON_ARRA...	1 row(s) affected Rows matched: 1 Changed: 1 ...	0.031 sec
✓	21 23:19:06	SELECT book_id, title, embedding FROM ...	10 row(s) returned	0.000 sec / 0.000 sec
✓	22 23:20:07	SELECT book_id, title, author, genre, ...	3 row(s) returned	0.015 sec / 0.000 sec
✓	23 23:21:08	SELECT book_id, title, author, genre, ...	3 row(s) returned	0.000 sec / 0.000 sec

Object Info

Session

106

ORDER BY book_id;

107

108

ALTER TABLE Books

109

ADD COLUMN embedding JSON;

Output

Action Output

#	Time	Action	Message	Duration / Fetch
⚠	1 23:10:45	DROP DATABASE IF EXISTS bookflow_db	0 row(s) affected, 1 warning(s): 1008 Can't drop dat...	0.015 sec
✓	2 23:10:45	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
✓	3 23:10:45	USE bookflow_db	0 row(s) affected	0.000 sec
✓	4 23:11:16	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓	5 23:12:31	CREATE TABLE Books (book_id INT AUTO_IN...	0 row(s) affected	0.031 sec
✓	6 23:13:00	DESCRIBE Books	6 row(s) returned	0.015 sec / 0.000 sec
✓	7 23:14:12	INSERT INTO Books (title, author, genre, ...	10 row(s) affected Records: 10 Duplicates: 0 War...	0.016 sec
✓	8 23:14:58	SELECT book_id, title, author, genre, p...	10 row(s) returned	0.000 sec / 0.000 sec
✓	9 23:15:28	ALTER TABLE Books ADD COLUMN embedding J...	0 row(s) affected Records: 0 Duplicates: 0 Warnin...	0.032 sec

